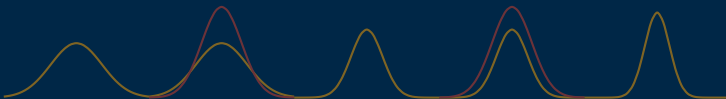


Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

Masters dissertation ▷ Thiago Tomás de Paula



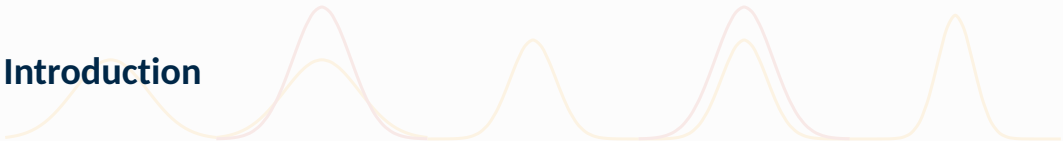
August 4th, 2026

Graduate Program in Mechanical Engineering

University of Brasília



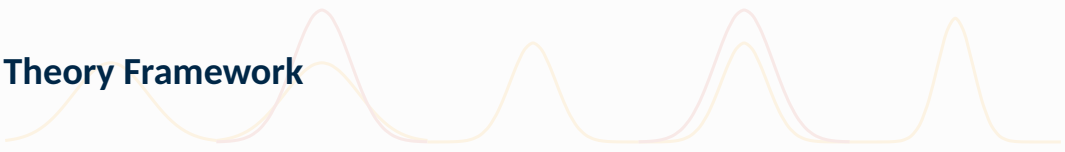
Introduction





- Contextualize the dissertation within the graduate program
- List lectures attended, papers published, Axia software
- Comment the 3-phase structure of the defense
- Contextualize the dissertation's theme
- Present the inverse problem
- Illustrate the effects the work is after
- Literature review: model-space methods
- Literature review: data-space (i.e., outline research gap)
- Lay down research questions
- Layout the remainder of the presentation

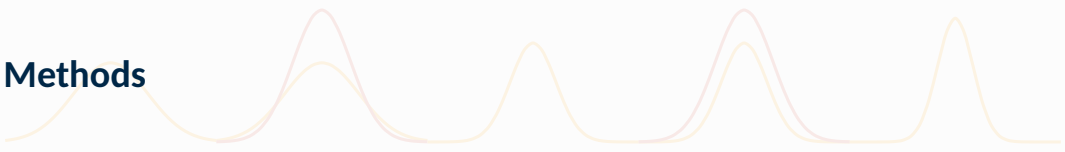
Theory Framework

The background of the slide features a series of overlapping, wavy lines in a light yellow and a soft pink color, creating a decorative, flowing pattern across the middle of the page.



- Impose notation and concepts - the history and validation periods
- Impose notation and concepts - the structure of data
- Impose notation and concepts - the prediction forward model
- Statistical framework - present Bayes theorem
- Statistical framework - "1D" example of Bayes theorem
- Statistical framework - example of Bayes theorem on PDFs
- Statistical framework - state the MDA approach (compare with ES)
- Derive the base DSI-ESMDA equation
- Comments on limitations
- Localization of covariance as a way of mitigating the effects of small ensembles (picture on anisotropy)
- Derive the Gated DSI-ESMDA equation (figure illustrating the idea behind the gating)
- Summarize and compare the derived equations (make a note on the lack of observation perturbation)

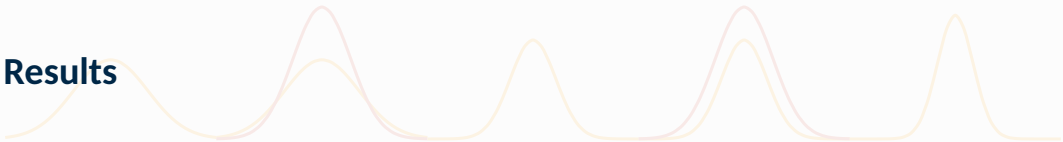
Methods

A decorative graphic consisting of two overlapping wavy lines, one yellow and one pink, that span the width of the slide. The lines are smooth and rhythmic, with the pink line generally positioned slightly above the yellow line.



- Present the case studies - petrophysical properties of each reservoir
- Present the case studies - (normalized) field oil production total of each reservoir
- Present the workflow behind the experiments (including noise modeling)
- Present the evaluation metrics used (compare meanings)
- Present the time, localization and features parameters employed

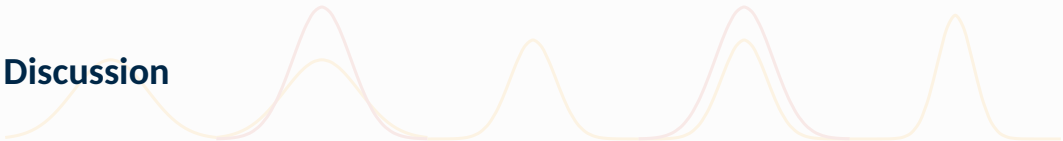
Results





- Present the results attained under each case study (minimal commentary)

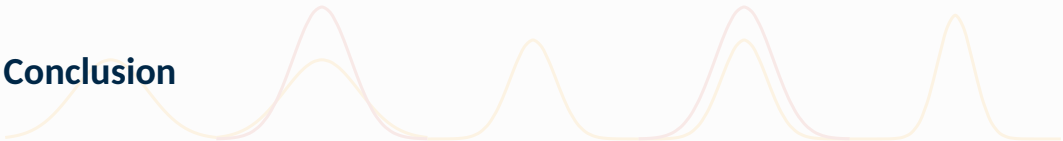
Discussion





- Summarize key findings
- Analyze (interpret) each main finding
- Summarize the analysis

Conclusion





- Recap the work: its context within the degree and as a research effort, its gap.
- Recap the main findings and their analysis
- State practical implications
- Comment on limitations and future research
- Make a closing statement (answer research questions)

Gated DSI-ESMDA

Reducing variance loss in posteriors

Thiago Tomás de Paula

Graduate students at University of Brasília, Brazil

August 4th, 2026

“It’s hard to make predictions, specially about the future”

— Yogi Berra

Thank you!